



Kinetic modeling study of toluene pyrolysis at low pressure

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ABSTRACT

A detailed kinetic model, consisting of 137 species and 530 reactions, was developed to simulate toluene pyrolysis at low pressure within the temperature range from 1270 to 1870 K. The mole fraction profiles predicted for pyrolysis species up to phenanthrene were in good agreement with the experiment. The decomposition pathways of toluene and the growth pathways to polycyclic aromatic hydrocarbons (PAHs) were discussed from reaction flux analysis. Toluene decomposes through the reaction sequence $C_6H_5CH_3 \rightarrow C_6H_5CH_2 \rightarrow C_7H_6 \rightarrow c-C_5H_5 \rightarrow C_3H_3$, which also has a predominant contribution to the production of acetylene. Furthermore, sensitivity analysis showed that the primary decomposition reactions of toluene, $C_6H_5CH_3 = C_6H_5CH_2 + H$ and $C_6H_5CH_3 = C_6H_5 + CH_3$, have great influences on the formation of small molecules, such as phenyl radical, benzyl radical, C2- and C3-species, which are critical to the formation of PAHs in the pyrolysis of toluene.

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1. Introduction

Toluene is a major aromatic component in practical fuels and also a desirable fuel additive due to its high energy density and high anti-knock rating [1–4]. Not only is toluene one of the simplest derivatives of benzene, its concentration is usually large in fuel-rich combustion processes, and its chemistry is a foundation for large aromatics formation. These considerations make its combustion an ideal system to study the formation mechanism of PAHs and soot. Indeed, toluene combustion has been extensively studied using a variety of experimental techniques [5–30]. Among these studies, toluene pyrolysis is of special interest because it exhibits the chemistry related to both fuel decomposition and PAH growth.

Some kinetic modeling studies of oxidation and pyrolysis of toluene were proposed for comparison with experimental results. In 1970s, McLain et al. proposed a simple kinetic model, which included global reactions of phenyl and benzyl radicals with O_2 , to simulate the oxidation of toluene [31]. In addition, a detailed mechanism was developed to study toluene oxidation by Bittker [32]. Studies on developing the toluene mechanism in the past two decades are listed in Table 1. Recently, we reported a new experimental investigation of low-pressure toluene pyrolysis in the temperature range of 1270–1870 K using the synchrotron vacuum ultraviolet photoionization mass spectrometry [33]. Comprehensive identification and concentration measurements were

made for the pyrolysis products and intermediates including isomers and radicals, which provide detailed chemical information for the development of kinetic models. The purpose of this work is to develop a detailed kinetic model for toluene pyrolysis based on validation against this new set of experimental data. The proposed model is critical to further development of mechanisms for toluene combustion.

2. Kinetic modeling

The chemical kinetic mechanism was mainly constructed from four previous mechanisms. It consists of 137 species and 530 elementary reactions. Table 2 lists selected reactions and their rate constants. The full mechanism is included in the supplementary content. For the C0–C6 submechanism, the USC Mech V2.0 [34] developed by Wang et al. was used, which is based on GRI-Mech and the kinetic models of H_2/CO [35], ethylene [36], acetylene [37], C3 fuel [38,39], and 1,3-butadiene [40] combustion developed in subsequent studies. Reactions from the H-abstraction C_2H_2 -addition (HACA) mechanism proposed by Wang and Frenklach [41], soot formation mechanism presented by Appel et al. (ABF mechanism) [42] and low-pressure soot model developed by Richter et al. (MIT mechanism) [43] were assembled as the sub-mechanisms of aromatics growth. Furthermore, many newly studied reactions were added to this model; and rate constants of some reactions in above mechanisms were updated with recent calculated or experimental results. Detailed discussions of the reactions of toluene decomposition, benzyl decomposition, small molecule reactions with C0–C6 and PAH reactions with C8–C14 are given below.

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Nomenclature

$p\text{-C}_3\text{H}_4$	propyne	$\alpha\text{-C}_3\text{H}_4$	allene
$c\text{-C}_5\text{H}_5$	cyclopentadienyl radical	$l\text{-C}_5\text{H}_5$	2-penten-4-ynyl radical
$o\text{-C}_6\text{H}_4$	benzyne	$l\text{-C}_6\text{H}_4$	3-hexene-1,5-diyne
A2	naphthalene	A2-1	1-naphthyl radical
A2R5	acenaphthylene	A3	phenanthrene
A3L	anthracene	P2	biphenyl
P2-	biphenyl radical		

2.1. Decomposition of toluene

The primary pyrolysis pathways of toluene are H and CH_3 scission reactions, shown as reactions R1 and R2.



There are several experimental and theoretical studies of these two competing reaction pathways. Most of them indicate that R1 is the predominant channel [11,27,28]. In 2007, Klippenstein et al. calculated the rate constant of the reverse reactions of R1 and R2 using high-level ab initio methods with variable-reaction-coordinate transition-state theory [44]. Recently, Oehlschlaeger et al. reported rate constants of R1 and R2 from 1400 K to 1780 K by using ultraviolet (UV) absorption spectroscopy behind shock waves [17]. These rates [17] were further reduced by the fall-off curve calculated by Klippenstein et al. [44] and used in the present model.

The radicals H and CH_3 formed from the reactions R1 and R2 can attack toluene, which accelerate the consumption of toluene.



Reaction R3 has been studied by atomic resonance absorption spectroscopy (ARAS) [27,29,45], gas chromatograph [46,47], time-resolved UV lamp absorption [9] and UV laser absorption [14]. The rate constant of R3 obtained by the recent shock tube experiment using UV laser absorption of the benzyl radical at 266 nm [14] was adopted in our model. The rate constants of reactions R4 and R5 from Brezinsky et al. [15] and Wang et al. [34] were also included in the model.

2.2. Decomposition of the benzyl radical

There is still no general consensus about the decomposition pathways of the benzyl radical, although some related experimental and theoretical studies have been reported. Colket and Seery [8] studied the decomposition reactions of the benzyl radical to form

$c\text{-C}_5\text{H}_5$ (cyclopentadienyl) + C_2H_2 and $\text{C}_4\text{H}_4 + \text{C}_3\text{H}_3$, which have been used extensively in many models [10,16,18]. In 1997, Jones et al. calculated four possible decomposition pathways, including a ring-opening pathway to an open-chain $\text{C}_7\text{H}_6 + \text{H}$ and three separate pathways to $c\text{-C}_5\text{H}_5 + \text{C}_2\text{H}_2$ using ab initio methods [48]. Among them, the benzyl radical produces $\text{C}_7\text{H}_6 + \text{H}$ via the direct ring-opening reaction and $c\text{-C}_5\text{H}_5 + \text{C}_2\text{H}_2$ via a 6-methylenebicyclo[3.1.0]hex-3-en-2-yl (MBH) intermediate are two dominant reaction channels. However, several experimental and theoretical studies concluded that the H scission from the benzyl radical may be a dominant channel in the decomposition of the benzyl radical [9,11,25,49–57]. A series of shock tube experiments using time-resolved UV absorption by Troe et al. [9,25,49–51] indicated that the benzyl radical decomposes to C_7H_6 (R7), which was also supported by ARAS observations [11,52,53].



The rate constant of R7 was obtained by Oehlschlaeger et al. using shock tube experiments with UV laser absorption at 266 nm [54]. They also suggested that the $c\text{-C}_5\text{H}_5$ and C_2H_2 observed in Colket and Seery's experiments [8] were most likely formed through further reaction of $\text{C}_7\text{H}_6 + \text{H}$ rather than the direct decomposition of the benzyl radical [54]. Recently, C_7H_6 was proposed to be fulvenallene by Cavallotti et al. [55] and da Silva et al. [56], which has been observed as the dominant C_7H_6 species in both fuel-rich premixed flame [19] and pyrolysis [33] experiments of toluene. In this model, the rate constants of R7 is taken from Oehlschlaeger et al. [54], and further reduced by the pressure dependence calculated by da Silva et al. [56]. The reactivity of fulvenallene is still unclear. Cavallotti et al. [55] computed the reaction of fulvenallene with hydrogen atoms and obtained the rate constant of reaction R10. However, the authors pointed out that the rate was underestimated. Detilleux and Vandooen [58] fitted the pre-exponential of R10 to a value that can predict the mole fraction of cyclopentadiene observed in their toluene flame, this rate was adopted for the present model.

Table 1
Previous studies on developing the toluene mechanism.

Year	Temperature range (K)	Pressure	Reaction type	Reactor	Ref.
1992	1100–1190	1 atm	Oxidation	Flow reactor	[7]
1994	1200–1850	10 atm	Pyrolysis	Shock tube	[8]
2002	1000–1375	1 atm	Oxidation	Jet-stirred reactor	[10]
2004	1200–1500	25, 50, 610 bar	Oxidation	Shock tube	[12]
2005	873–923	1 bar	Oxidation	Flow stirred tank	[13]
		8.7 bar	Oxidation	Shock tube	
2006	1200–1900	27, 45 bar	Pyrolysis	Shock tube	[15,16]
2008	700–1300	1–5 MPa	Oxidation	Shock tube	[18]
2009	700–1800	36 Torr	Flame	Premixed laminar flame	[58]
2009	1021–1400	10–61 atm	Oxidation	Shock tube	[21]

Table 2The rate constants for the selected reactions for the toluene pyrolysis, $k = AT^n \exp(-E/RT)$. The units are in cm^3 , mol, s, cal.

Selected reactions	A	n	E	Ref.
<i>Selected reactions of C7 species</i>				
1 $\text{C}_6\text{H}_5\text{CH}_3 = \text{C}_6\text{H}_5\text{CH}_2 + \text{H}$	2.83×10^{64}	-13.99	120,613	See text
2 $\text{C}_6\text{H}_5\text{CH}_3 = \text{C}_6\text{H}_5 + \text{CH}_3$	2.16×10^{88}	-20.61	144,292	See text
3 $\text{C}_6\text{H}_5\text{CH}_3 + \text{H} = \text{C}_6\text{H}_5\text{CH}_2 + \text{H}_2$	6.47×10^{90}	3.98	3394	[14]
4 $\text{C}_6\text{H}_5\text{CH}_3 + \text{H} = \text{C}_6\text{H}_6 + \text{CH}_3$	1.56×10^{13}	0.0	5796	[62]
5 $\text{C}_6\text{H}_5\text{CH}_3 + \text{CH}_3 = \text{C}_6\text{H}_5\text{CH}_2 + \text{CH}_4$	3.16×10^{11}	0.0	9500.0	[34]
6 $\text{C}_6\text{H}_5\text{CH}_2 + \text{H} = \text{C}_6\text{H}_5 + \text{CH}_3$	4.50×10^{58}	-11.9	51,860	[44]
7 $\text{C}_6\text{H}_5\text{CH}_2 = \text{C}_7\text{H}_6 + \text{H}$	4.38×10^{13}	0.078	75,900	See text
8 $\text{C}_6\text{H}_5\text{CH}_2 + \text{H} = \text{C}_7\text{H}_6 + \text{H}_2$	5.00×10^{13}	0.0	0.0	Est.
9 $\text{C}_6\text{H}_5\text{CH}_2 = c\text{-C}_5\text{H}_5 + \text{C}_2\text{H}_2$	3.00×10^{13}	0.0	70,000	[8] ^a
10 $\text{C}_7\text{H}_6 + \text{H} \rightarrow c\text{-C}_5\text{H}_5 + \text{C}_2\text{H}_2$	3.00×10^{14}	0.0	0.0	[58]
11 $\text{C}_7\text{H}_5 + \text{H} = \text{C}_7\text{H}_6$	2.00×10^{12}	0.0	0.0	Est.
12 $\text{C}_7\text{H}_6 + \text{H} = \text{C}_7\text{H}_5 + \text{H}_2$	1.90×10^{08}	1.847	4965	[59]
13 $\text{C}_7\text{H}_6 + \text{CH}_3 = \text{C}_7\text{H}_5 + \text{CH}_4$	1.87×10^{04}	2.724	6008	[59]
14 $\text{C}_7\text{H}_5(+\text{M}) = \text{C}_4\text{H}_2 + \text{C}_3\text{H}_3(+\text{M})$	3.00×10^{12}	-0.075	62,300	Est. ^b
Low pressure limit	1.00×10^{45}	8.4	47,500	
15 $\text{C}_7\text{H}_5 = \text{C}_5\text{H}_3 + \text{C}_2\text{H}_2$	2.00×10^{11}	-0.075	62,300	Est. ^b
Low pressure limit	1.00×10^{45}	8.4	47,500	
16 $\text{C}_7\text{H}_5 + \text{H} = \text{C}_5\text{H}_4 + \text{C}_2\text{H}_2$	1.00×10^{14}	0.0	0.0	Est.
<i>Selected reactions of C6 species</i>				
17 $a\text{-C}_3\text{H}_4 + \text{C}_3\text{H}_3 = \text{C}_6\text{H}_6 + \text{H}$	2.20×10^{11}	0.0	2000	[96]
18 $i\text{-C}_4\text{H}_5 + \text{C}_2\text{H}_2 = \text{C}_6\text{H}_6 + \text{H}$	3.00×10^{11}	0.0	14,900	[96]
19 $\text{C}_3\text{H}_3 + \text{C}_3\text{H}_3 = \text{C}_6\text{H}_6$	1.64×10^{66}	-15.9	27,529	[64]
Duplicate	1.20×10^{35}	-7.4	5058	
20 $\text{C}_3\text{H}_3 + \text{C}_3\text{H}_3 = \text{fulvene}$	7.25×10^{65}	-16.0	25,035	[64]
Duplicate	4.19×10^{39}	-9.0	6098	
21 $\text{C}_4\text{H}_4 + \text{C}_2\text{H}_2 = \text{C}_6\text{H}_6$	4.47×10^{11}	0.0	30,010	[43]
22 $\text{C}_6\text{H}_5 + \text{H} = \text{C}_6\text{H}_6$	8.02×10^{19}	-2.0	1968.0	[97]
23 $\text{C}_6\text{H}_5 + \text{H}_2 = \text{C}_6\text{H}_6 + \text{H}$	5.71×10^{04}	2.43	6273.00	[34]
24 $\text{C}_3\text{H}_3 + \text{C}_3\text{H}_3 = \text{C}_6\text{H}_5 + \text{H}$	2.20×10^{33}	-6.0	15,940	[98]
25 $\text{C}_6\text{H}_5 = o\text{-C}_6\text{H}_4 + \text{H}$	3.00×10^{13}	0.0	72,923	See text
26 $\text{C}_6\text{H}_5 + \text{H} = o\text{-C}_6\text{H}_4 + \text{H}_2$	2.00×10^{11}	1.1	24,500	[34]
27 $l\text{-C}_6\text{H}_4 + \text{H} = \text{C}_6\text{H}_5$	3.60×10^{77}	-20.1	28,100	[34]
28 $n\text{-C}_4\text{H}_3 + \text{C}_2\text{H}_2 = \text{C}_6\text{H}_5$	1.40×10^{67}	-17.4	23,000	[34]
29 $\text{C}_3\text{H}_2 + \text{C}_3\text{H}_2 = l\text{-C}_6\text{H}_4$	3.00×10^{11}	0.0	0.0	Est. ^c
30 $o\text{-C}_6\text{H}_4 = \text{C}_4\text{H}_2 + \text{C}_2\text{H}_2$	2.00×10^{14}	0.0	83,000	See text
31 $\text{C}_4\text{H}_2 + \text{C}_2\text{H}_2 = l\text{-C}_6\text{H}_4$	4.47×10^{11}	0.0	30,010	Est. ^d
32 $l\text{-C}_6\text{H}_4 + \text{H} = o\text{-C}_6\text{H}_4 + \text{H}$	2.20×10^{47}	-9.98	24,000	[34]
33 $\text{C}_6\text{H}_3 + \text{H} = l\text{-C}_6\text{H}_4$	4.20×10^{44}	-10.3	7890	[34]
34 $l\text{-C}_6\text{H}_4 + \text{H} = \text{C}_6\text{H}_3 + \text{H}_2$	1.33×10^{06}	2.53	9240	[34]
35 $\text{C}_6\text{H}_3 + \text{H} = \text{C}_4\text{H}_2 + \text{C}_2\text{H}_2$	2.40×10^{19}	-1.60	2800	[34]
36 $\text{C}_6\text{H}_3 + \text{H} = \text{C}_6\text{H}_2 + \text{H}_2$	3.00×10^{13}	0.0	0.0	[34]
37 $\text{C}_3\text{H}_2 + \text{C}_3\text{H}_2 = \text{C}_6\text{H}_2 + \text{H}_2$	7.20×10^{11}	0.0	0.0	Est. ^c
<i>Selected reactions of C5 species</i>				
38 $\text{C}_5\text{H}_6(+\text{M}) = c\text{-C}_5\text{H}_5 + \text{H}(+\text{M})$	7.65×10^{19}	-1.18	84,473	[58]
Low pressure limit	1.07×10^{80}	-16.79	16,985	
39 $\text{C}_5\text{H}_6 + \text{C}_6\text{H}_5 = c\text{-C}_5\text{H}_5 + \text{C}_6\text{H}_6$	3.10×10^{11}	0.0	5500.0	[99]
40 $\text{C}_5\text{H}_6 + \text{CH}_3 = c\text{-C}_5\text{H}_5 + \text{CH}_4$	5.00×10^{11}	0.0	5500.0	[65]
41 $c\text{-C}_5\text{H}_5 = \text{C}_3\text{H}_3 + \text{C}_2\text{H}_2$	3.00×10^{12}	-0.075	62,300	[66]
Low pressure limit	1.00×10^{45}	8.4	47,500	
42 $\text{C}_3\text{H}_3 + \text{C}_2\text{H}_2 = \text{C}_5\text{H}_4 + \text{H}$	2.35E11	0.0	9995.2	Est. ^e
43 $\text{C}_5\text{H}_3 + \text{H} = \text{C}_5\text{H}_4$	1.0×10^{13}	0.0	0.0	[100]
<i>Selected reactions of C0–C4 species</i>				
44 $\text{H} + \text{H} + \text{M} = \text{H}_2 + \text{M}$	1.78×10^{18}	-1.0	0.0	[101]
Third body efficiencies: $\text{H}_2 = 0.0$, $\text{AR} = 0.63$				
45 $\text{CH} + \text{H}_2 = \text{CH}_2 + \text{H}$	1.11×10^{08}	1.79	1670	[101]
46 $\text{CH}_2 + \text{CH} = \text{C}_2\text{H}_2 + \text{H}$	4.0×10^{13}	0.0	0.0	[101]
47 $\text{CH}_2 + \text{CH}_2 = \text{C}_2\text{H}_2 + \text{H}_2$	3.2×10^{13}	0.0	0.0	[101]
48 $\text{CH}_4 + \text{H} = \text{CH}_3 + \text{H}_2$	6.6×10^{08}	1.62	10,840	[101]
49 $\text{CH}_3 + \text{H}(+\text{M}) = \text{CH}_4(+\text{M})$	1.27×10^{16}	-0.63	383.0	[101]
Low pressure limit	2.48×10^{33}	-4.76	2440	
Troe parameters: 0.7830, 74.00, 2941.0, 6964.0				
Third body efficiencies:				
$\text{H}_2 = 2.0$, $\text{CH}_4 = 2.0$, $\text{C}_2\text{H}_6 = 3.0$, $\text{AR} = 0.7$				
50 $\text{C}_3\text{H}_2 + \text{CH}_3 = \text{C}_4\text{H}_4 + \text{H}$	1.00×10^{13}	0.0	0.0	[34]
51 $p\text{-C}_3\text{H}_4 = a\text{-C}_3\text{H}_4$	6.40×10^{61}	-14.6	88,200	[34]
52 $\text{C}_3\text{H}_3 + \text{H} = \text{C}_3\text{H}_2 + \text{H}_2$	5.00×10^{13}	0.0	1000	[34]
53 $\text{C}_3\text{H}_3 + \text{H} = p\text{-C}_3\text{H}_4$	7.00×10^{11}	0.0	0.0	[34]
54 $\text{C}_3\text{H}_3 + \text{H} = a\text{-C}_3\text{H}_4$	1.50×10^{12}	0.0	0.0	[34]
55 $\text{C}_3\text{H}_3 + \text{CH}_2 = \text{C}_4\text{H}_4 + \text{H}$	5.00×10^{13}	0.0	0.0	[34]
56 $p\text{-C}_3\text{H}_4 + \text{H} = a\text{-C}_3\text{H}_4 + \text{H}$	2.30×10^{15}	-0.3	7600	[34]

Table 2 (continued)

Selected reactions	A	n	E	Ref.
<i>Reactions of C8 species</i>				
57 $C_6H_6 + C_2H = C_6H_5C_2H + H$	1.00×10^{13}	0.0	0.0	[42]
58 $C_6H_5CH_3 + CH_2 = C_6H_5C_2H_3 + H_2$	1.00×10^{13}	0.0	0.0	Est. ^g
59 $C_6H_5CH_2 + CH_3 = C_6H_5C_2H_5$	2.00×10^{13}	0.0	0.0	Est. ^g
60 $C_6H_5 + C_4H_4 = C_6H_5C_2H_3 + C_2H$	3.20×10^{11}	0.0	1900	[43]
61 $C_6H_5 + C_4H_6 = C_6H_5C_2H_3 + C_2H_3$	3.20×10^{11}	0.0	1900	[43]
62 $C_4H_4 + C_4H_4 = C_6H_5C_2H_3$	1.50×10^{14}	0.0	38,000	[43]
63 $C_6H_5C_2H_5 + H = C_6H_6 + C_2H_5$	1.20×10^{13}	0.0	5100	[43]
64 $C_6H_5C_2H_5 = C_6H_5C_2H_3 + H_2$	5.01×10^{12}	0.0	64,000	[43]
65 $C_6H_5 + C_2H_6 = C_6H_5C_2H_5 + H$	2.51×10^{12}	0.0	6190	Est. ^f
<i>Reactions of C9 species</i>				
66 $C_7H_6 + C_2H_2 = C_9H_8$	1.44×10^{292}	−78.2	245,010	[87]
67 $C_6H_5CH_2 + C_2H_2 = C_9H_8 + H$	3.16×10^{04}	2.48	11,061	[22]
68 $C_6H_5 + C_3H_4 = C_9H_8 + H$	5.13×10^{02}	3.19	2788	[84]
69 $C_6H_5 + C_3H_3 = C_9H_8$	1.00×10^{75}	−17.8	39,600	Est. ^h
70 $C_6H_6 + C_3H_3 = C_9H_8 + H$	6.26×10^{09}	2.61	56,500	Est.
71 $C_9H_8 = C_9H_7 + H$	1.73×10^{68}	−15.2	116,372	[22]
72 $C_9H_7 + C_3H_3 = A2R5 + H2$	4.10×10^{43}	−9.20	15,153	[22]
73 $C_9H_7 + C_3H_2 = A2R5 + H$	1.00×10^{13}	0.0	0.0	Est.
74 $C_9H_7 + c-C_5H_5 = A3 + 2 H$	6.39×10^{29}	−4.03	35,205	[22]
<i>Reactions of C10 species</i>				
75 $C_6H_4C_2H + C_2H = C_6H_5(C_2H)_2$	5.00×10^{13}	0.0	0.0	[41]
76 $C_6H_4C_2H + C_2H_2 = C_6H_5(C_2H)_2 + H$	5.50×10^{32}	−5.5	27,600	[41]
77 $C_6H_5(C_2H)_2 + H = A2-1$	2.00×10^{82}	−20.2	36,900	[41]
78 $C_6H_5CH_2 + C_3H_3 = A2 + 2 H$	2.00×10^{14}	0.0	20,000	Est.
79 $c-C_5H_5 + c-C_5H_5 = A2 + 2 H$	6.39×10^{29}	−4.03	35,205	[22]
80 $C_6H_5 + n-C_4H_3 = A2$	1.51×10^{75}	−17.8	39,600	[43]
81 $C_6H_5 + n-C_4H_3 = A2-1 + H$	1.84×10^{75}	−16.1	57,630	[43]
82 $C_6H_5 + n-C_4H_3 = A2-2 + H$	1.84×10^{72}	−16.1	57,630	[43]
83 $C_9H_7 + CH_2 = C_{10}H_9$	1.00×10^{11}	0.0	0.0	[92]
84 $n-A1C_2H_2 + C_2H_2 = A2 + H$	2.10×10^{15}	−1.07	4800	[41]
85 $A1C2H3 + C_2H_2 = A2 + H$	2.10×10^{15}	−1.07	6000	[41]
86 $C_6H_5CH_2 + C_3H_3 = C_{10}H_{10}$	1.51×10^{75}	−17.8	39,600	Est. ^h
87 $C_6H_5CH_3 + C_3H_2 = C_{10}H_{10}$	1.51×10^{75}	−17.8	39,600	Est. ^h
88 $C_{10}H_9 + H = C_{10}H_{10}$	1.00×10^{13}	0.0	0.0	Est.
89 $C_{10}H_9 + H = A2 + H_2$	1.00×10^{13}	0.0	0.0	Est.
<i>Reactions of C12–C14 species</i>				
90 $o-C_6H_4 + o-C_6H_4 = \text{biphenylene}$	4.60×10^{12}	0.0	0.0	[43]
91 $C_6H_5 + C_6H_5 = \text{biphenylene} + H_2$	5.60×10^{12}	−0.07	7550	Est. ⁱ
92 $C_6H_5 + C_6H_5 = P2$	1.14×10^{13}	0.0	0.0	[102]
93 $C_6H_6 + C_6H_5 = P2 + H$	5.60×10^{12}	−0.07	7550	[41]
94 $P2 = P2- + H$	9.00×10^{37}	−6.63	119,580	[41]
95 $P2 + H = P2- + H$	2.50×10^{14}	0.0	16,000	[41]
96 $P2- = \text{biphenylene} + H$	1.30×10^{16}	0.0	33,203	[43]
97 $P2- + H = \text{biphenylene} + H_2$	6.02×10^{12}	0.0	0.0	[43]
98 $C_6H_5CH_2 + C_6H_5 = C_{13}H_{12}$	6.00×10^{12}	0.0	220	[43]
99 $C_{13}H_{12} + H = C_{13}H_{11} + H_2$	3.23×10^{09}	2.10	15,842	[43]
100 $C_{13}H_{11} + H = C_{13}H_{12}$	1.00×10^{13}	0.0	0.0	Est.
101 $C_{13}H_{10} + H = C_{13}H_{11}$	3.20×10^{51}	−12.6	13,500	Est. ^j
102 $C_{13}H_{11} + H = C_{13}H_{10} + H_2$	1.00×10^{13}	0.0	0.0	Est.
103 $C_{13}H_{10} + H = C_{13}H_9 + H_2$	1.00×10^{13}	0.0	0.0	Est.
104 $C_6H_5CH_2 + C_6H_5CH_2 \rightarrow C_{14}H_{14}$	5.00×10^{12}	0.0	454	[103]
105 $C_{14}H_{14} = C_{14}H_{12} + H_2$	1.50×10^{15}	0.0	84,700	Est. ^k
106 $C_{14}H_{12} = A3 + H_2$	2.50×10^{15}	0.0	84,700	Est. ^k
107 $P2- + C_2H_2 = A3 + H$	4.60×10^{06}	1.97	7300	[41]
108 $A3L \rightarrow A3$	7.94×10^{12}	0.0	65,000	[8]
Reverse	5.37×10^{12}	0.0	67,000	
109 $C_{14}H_{14} + H = C_{14}H_{12} + H + H_2$	8.00×10^{13}	0.0	8235	Est. ^l

^a Reduced by a factor of 10 considering the pressure fall-off.^b Refer to the reaction $c-C_5H_5 = C_3H_3 + C_2H_2$.^c Refer to the reaction $C_3H_3 + C_3H_3 = C_6H_6$ [104].^d Refer to the reaction $C_4H_4 + C_2H_2 = C_6H_6$ [43].^e Refer to the reaction $C_3H_3 + C_2H_2 = c-C_5H_5$ [68].^f Refer to the reaction $C_6H_5 + C_2H_4 = C_6H_5C_2H_3 + H$ [41].^g Refer to the reaction $C_6H_6 + C_2H = C_6H_5C_2H + H$ [41].^h Refer to the reaction $C_6H_5 + n-C_4H_3 = A2$ [43].ⁱ Refer to the reaction $C_6H_6 + C_6H_5 = P2 + H$ [41].^j Refer to the reaction $C_6H_6 + H = c-C_6H_7$ [42].^k Refer to the reaction $C_4H_6 = C_4H_4 + H_2$ [43].^l Refer to the reaction $C_6H_5C_2H_5 + H = C_6H_5C_2H_3 + H_2 + H$ [43].



Cavallotti et al. [55] also suggested that a fulvenallene decomposition mechanism was probably active in addition to R10. More recently, the H abstraction from fulvenallene by H and CH₃ yielding fulvenallenyl radical (C₇H₅, R12 and R13) was studied by da Silva and Bozzelli [59]. The stabilization of C₇H₅ indicates that it is a possible decomposition product of fulvenallene. Reaction R14 and R15 are proposed to be the decomposition pathways of C₇H₅ in the present model, which is analogous to the decomposition channel of c-C₅H₅.

2.3. Small molecule reactions with C0–C6

C0–C6 species are mainly formed by the primary and the subsequent reactions in toluene pyrolysis. USC Mech V2.0 [34] was used as the basis for small molecule chemistry in this model, excluding reactions containing oxygen.

Recently, it has been suggested that the phenyl radical decomposes to o-C₆H₄ (benzyne) + H rather than n-C₄H₃ + C₂H₂ [60–62]. The reaction rate constant for R25 suggested by Baulch et al. [62] was adopted in this model with reduction by the fall-off curve calculated by Madden et al. [60]. Reaction R30 was studied by Xu et al. [63] using a shock tube experiment at pressure between 1.4–2.0 bar. The rate of R30 obtained by Xu et al. was reduced by a factor of 6 due to fall-off and taken into the present model.



The propargyl recombination submechanism used in the present model is from the study on an allene and propyne flame at 25 Torr [64]. Some reactions related to C6-species from the ABF mechanism [42] were added to the model, as well as some C5-species reactions reported by Bacskey et al. [65] and Kern et al. [66] such as reactions R38–R40.



The reverse reaction of R41 is the decomposition pathway of the cyclopentadienyl radical, which was studied by several groups using both theory and experiment [66–69]. Its rate constant used in this model is from the pyrolysis of cyclopentadiene at 100–450 Torr studied by Kern et al. [66].

2.4. PAH reactions with C8–C14

The HACA mechanism [41], which interprets the ring growth of PAHs via C2 addition channels, contributes to the submechanism of PAH growth in this work. Growth pathways of aromatics by addition of C3-, C4-, C5-, C6- and C7-species are also included, most of which were collected from the MIT mechanism [43]. Furthermore, some H/H₂-elimination reactions (R99–R105) from

bicyclic aromatics such as diphenylmethane (C₁₃H₁₂) and bibenzyl (C₁₄H₁₄), yielding more stable tricyclic PAHs (fluorene C₁₃H₁₀ and phenanthrene C₁₄H₁₀), were used to simulate these PAHs. The rate parameters of many reactions involving PAHs are unknown. They were assigned based on analogous reactions.

The computation was performed using the CHEMKIN II program [70]. Most thermodynamic and transport data were available in the corresponding models, while other data were taken from the compilation of Burcat and Ruscic [71]. The experimental inlet condition was 1.24 vol.% toluene in argon [33]. The pressure in the middle of the flow reactor was used in our modeling, which was around 20 Torr as reported previously [33]. After considering the temperature error, the experimental temperatures were increased by 70 K to make the simulation result comparable with the experimental data.

3. Results and discussion

3.1. Soot formation

The contents of total carbon and hydrogen elements detected in the toluene pyrolysis experiment decreased as the temperature is increased. The missing carbon may be attributed to soot formation and/or measurement errors. The initial toluene concentration in the pyrolysis experiment is only 1.24%. Because one toluene molecule would decompose to about three molecules, the total mole number is increased slightly, by 2–3%. Hence, the total mole number is assumed to be constant. The content of carbon (hydrogen) element is the sum of mole fractions of all species detected in the experiment multiplied by its carbon (hydrogen) number. The contents of carbon and hydrogen element lost can be obtained by Eqs. (1) and (2), where M(C_xH_y) represents the mole fraction of C_xH_y, and 0.0124 is the mole fraction of the inlet toluene.

$$\text{C}_{\text{lost}} = 7 \times 0.0124 - \sum_{x,y} x \times \text{M}(\text{C}_x\text{H}_y) \quad (1)$$

$$\text{H}_{\text{lost}} = 8 \times 0.0124 - \sum_{x,y} y \times \text{M}(\text{C}_x\text{H}_y) \quad (2)$$

As shown in Fig. 1, H_{lost} has a bell-shaped profile with the peak value of 0.0342 at 1710 K while the total content of hydrogen element is 0.0992. Meanwhile, C_{lost} increases steadily as the temperature increases and reaches 0.049 at 1820 K, about one half of the

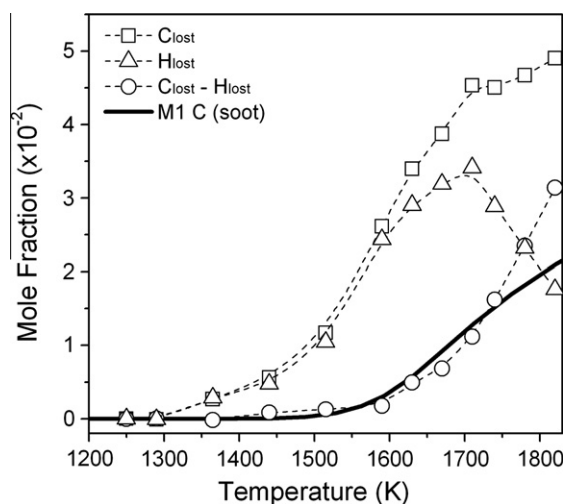


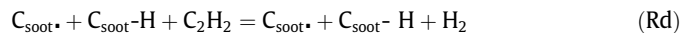
Fig. 1. Experimental and computed mole fraction profiles of soot. The open symbols with dash lines are experimental results [33] and the solid line is the modeling result by the model M1 with considering the soot formation.

total amount (0.0868). The quenching of radicals with the skimmer and uncertain photoionization cross sections of some species are possible reasons for H and C losses. However, because C_{lost} is much more than H_{lost} at high temperatures, the most plausible cause is solid soot formation. The experimentally detected largest PAH was $C_{16}H_{10}$ [33], and high PAHs larger than $C_{16}H_{10}$ were counted as the solid soot in the experiment. Indeed, we observed soot deposits on the wall of the alumina flow tube [33]. The atomic ratio of C to H in typical soot is about 7:1 [72]. The formation of soot would lead to loss of C with slightly influence on loss of H, which leads to result that C_{lost} is higher than H_{lost} . The estimated ratio of C_{lost} to H_{lost} caused by other factors is 1:1, because the C/H of most intermediates and products are around 1:1. This assumption is consistent with the experimental observations made below 1570 K, as shown in Fig. 1. The content of the lost carbon minus the lost hydrogen ($C_{\text{lost}} - H_{\text{lost}}$) would be indicative of the content of soot. As seen in Fig. 1, the soot content increases as the temperature increasing above 1570 K and reaches to 0.0314 at 1820 K. This analysis suggests that temperature for the onset of soot formation is about 1570 K, which is in good agreement with a previous study [73].

Several experimental and modeling studies of reactions of soot were reported previously [72,74–81]. Among them, reactions (Ra), (Rb) and (Rc) were proposed by Frenklach and Wang [81] to describe the surface reactions of soot by the addition of acetylene.



The rate constants of these three reactions were estimated based on analogous gas-phase reactions of one-ring aromatics in their work.



The reaction Rd is the combination of the reactions Ra and Rc, which implies acetylene is consumed by reacting with soot to produce H_2 + larger soot. To simplify the model, an overall reaction Re with fitted rate constants is proposed to predict the soot production that is equal to the carbon loss observed experimentally.



In this work, the model with reaction Re is named as Model 1 (M1), while the model without reaction Re is referred to as Model 2 (M2). M1 prediction for the soot content is in good agreement with the experimental data, as shown in Fig. 1. Both experimental and modeling data begin to increase at around 1570 K and reach about 0.025 at 1820 K. Furthermore, H_2 predicted by M2 is only 0.0120 at 1820 K, about one half of the experimental measurement (0.0228), while M1 predicts the concentration of H_2 much closer to the experimental value, as shown in Fig. 2a. The mole fractions of acetylene and polyynes up to $C_{10}H_2$ predicted by M2, as shown in Fig. 2, are about a factor of 2–3 times higher than the experimental data, while the mole fractions of $C_{2n}H_2$ ($n = 1-5$) predicted by M1 are in good agreement with the experimental data.

Therefore, we can conclude that the model predictions of quasi-equilibrium concentrations of polyynes are in better accordance with the experimental results after considering carbon deposition. The results will be discussed hereafter with the model M1.

3.2. Simulated results and pathway analysis

The mole fractions predicted for the decomposition products and PAHs up to phenanthrene are compared with those measured experimentally. The results are illustrated in Figs. 3 and 4. In general, the modeling results are in good agreement with the experi-

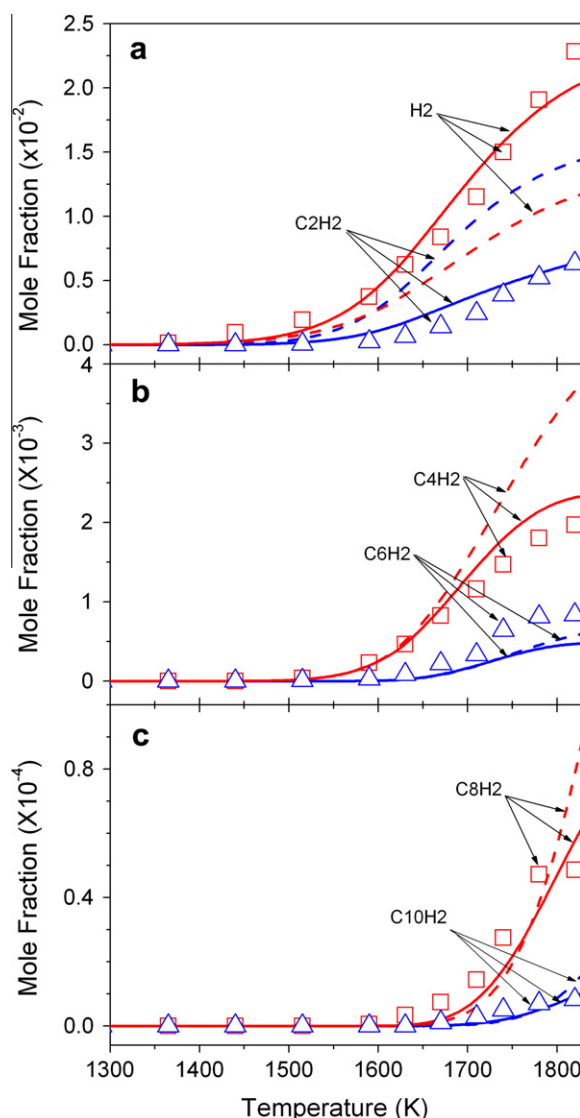


Fig. 2. Comparison between experimental and modeling predictions of mole fraction profiles of (a) H_2 and C_2H_2 , (b) C_4H_2 and C_6H_2 , and (c) C_8H_2 and $C_{10}H_2$. The experimental data are shown as open symbols. The solid and dash lines denote the modeling results by the model M1 with considering the soot formation and the model M2 without the soot formation, respectively.

ment. Based on reaction flux analysis at 1670 K, the production and consumption reactions for selected species are listed in Tables 3 and 4. Furthermore, the decomposition pathways of toluene are analyzed and shown in Fig 5; the growth pathways of PAHs in toluene pyrolysis at 1670 K are presented in Fig. 6. The thickness of the arrow reflects the flux of the pathway. Sensitivity analysis of selected species was performed and is illustrated in Figs. 7–11 to identify reaction channels with strong influences on these species.

3.2.1. Reactant and primary products

The predicted and experimental mole fraction profiles of four C7-species, C_7H_8 , C_7H_7 , C_7H_6 and C_7H_5 (5-ethynyl-1,3-cyclopentadien-5-yl radical), are compared and presented in Fig. 3a. Both experimental and computed mole fractions of toluene decrease following the same tendency. The complete disappearance occurs at 1820 K. As a critical intermediate species, C_7H_7 has a bell-shape mole fraction profile. The benzyl radical is produced from reactions R1, R3 and R5, and is consumed rapidly to form C_7H_6 and $c-C_5H_5$ at high temperatures. The temperature calculated for the maximum

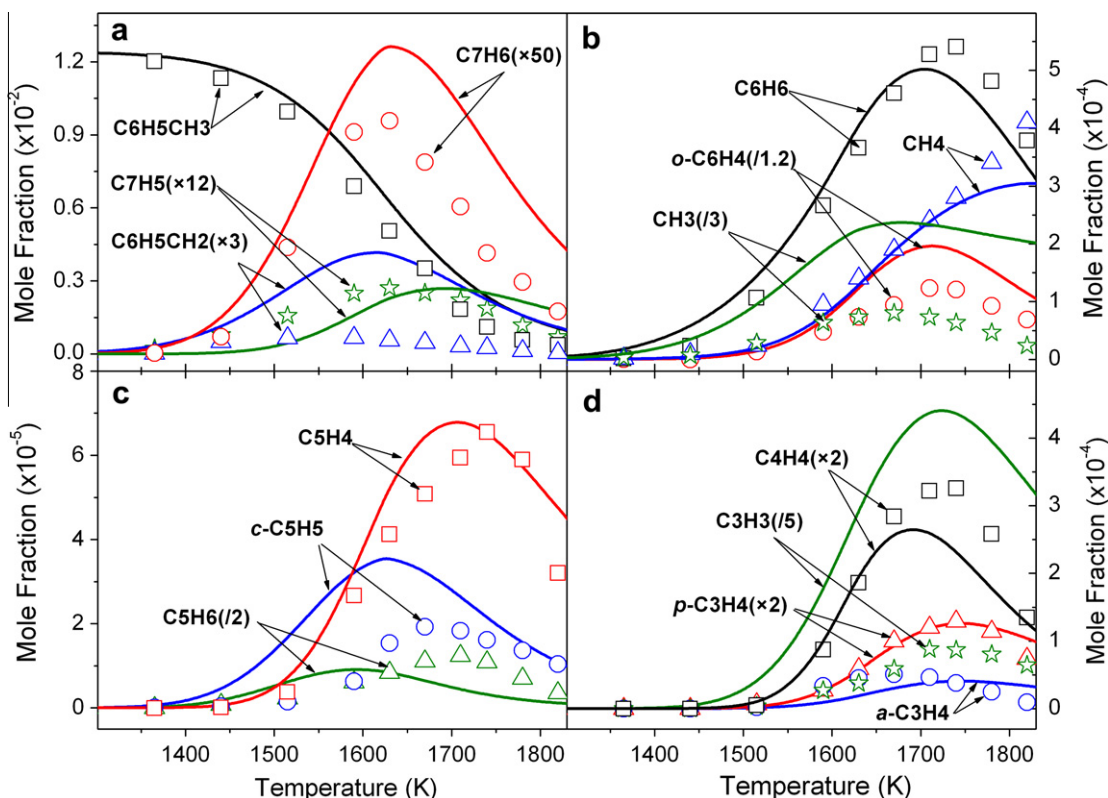


Fig. 3. Comparison between experimental and modeling predictions of mole fraction profiles of C1–C7-species. The experimental data are shown as open symbols. The solid lines denote the modeling results by the model M1.

mole fraction (T_M) of C_7H_7 is around 1620 K, which is close to the experimental value of 1590 K. However the maximum mole fraction (X_M) predicted by M1 is much higher than the experimental result. This may be partly caused by the quenching of the benzyl radical on the surface of the sampling nozzle.

Reaction flux analysis shows that about 75% of toluene decomposes to the benzyl radical through unimolecular decomposition (R1) and the attack of H atom (R3) at 1670 K. The destruction reactions (R1) and (R3) of toluene exhibits great influence on the formation of the benzyl radical, as shown in Fig. 7. The $C_6H_5CH_3 = C_6H_5 + CH_3$ pathway (R2) competes with reaction R1 and has an inhibiting effect on the production of the benzyl radical. The further destruction of the benzyl radical is dominated by unimolecular decomposition reactions, $C_6H_5CH_2 = c-C_5H_5 + C_2H_2$ (R9) and $C_6H_5CH_2 = C_7H_6 + H$ (R7). About 60% of the benzyl radical decomposes to C_7H_6 , while 12% produces $c-C_5H_5$. As presented in Fig. 7, the reactions of the benzyl radical forming C_7H_6 (R7) and (R8) and $c-C_5H_5$ (R9) play important roles in its consumption.

As shown in Fig. 3, the calculated mole fraction of C_7H_6 is in good agreement with the experiment in both shape and X_M . The reactivity of C_7H_6 is still unclear. Besides the $C_7H_6 + H \rightarrow c-C_5H_5 + C_2H_2$ pathway calculated by da Silva et al. [56] and Cavallotti et al. [55], C_7H_5 is a possible product of C_7H_6 . By the submechanism of C7-species list in Table 2, the model predicts a similar X_M of C_7H_5 to the experimental data, though the computed initial formation temperature (T_{Form}) is higher than experimental results. Detailed kinetic information of C_7H_6 and C_7H_5 is still needed.

Phenyl radical is another primary product of toluene pyrolysis. About 80% of the phenyl radical comes from the unimolecular decomposition of toluene (R2) and 13% from the attack of H (R6). The main products of the consumption of the phenyl radical are benzyne and C_3H_3 via the reactions $C_3H_3 + C_3H_3 = C_6H_5 + H$ (R24, 29% consumption) and $C_6H_5 = o-C_6H_4 + H$ (R25, 29% consumption).

Furthermore, reactions involving the phenyl radical can also produce benzene (R22), indene (R69), biphenyl (R91) and diphenylmethane (R98). Fig. 8 shows the sensitivity analysis for the phenyl radical. It can be seen that reaction R2 plays a dominant role in the formation of phenyl. The competing pathways (R1) and (R3) show inhibiting effects on phenyl formation. Because phenyl can react further and produce benzene (R22), propargyl radical (R24), benzyne (R25), indene (R69) and diphenylmethane (R98), these reactions will cause lower concentrations of the phenyl radical.

3.2.2. Small molecules

The experimental and computed mole fraction profiles of some selected small molecule species (C1–C6) are displayed in Fig. 3b–d. Fig. 3b presents the profiles of C_6H_6 , C_6H_4 , CH_4 , and CH_3 . The prediction of C_6H_6 , C_6H_4 and CH_4 are within the accepted error range. However, CH_3 is three times higher than the experimental observation, probably due to the quenching effect. Reaction flux analysis (Table 3) shows that reactions between toluene and H yielding benzene (R4), the recombination reactions of two propargyl radicals (R19), and the phenyl radical with H and H_2 (R22) and (R23) contribute 90% of benzene formation.

Two C_6H_4 isomers, 3-hexene-1,5-diyne (*l*- C_6H_4) and benzyne (*o*- C_6H_4), were detected in the experiment. The calculated mole fraction profile of *o*- C_6H_4 is about a factor of 30 higher than *l*- C_6H_4 . Further reaction flux analysis shows that most of the *o*- C_6H_4 is produced by the H elimination from the phenyl radical (R25), while *l*- C_6H_4 is formed from the recombination of C_3H_2 (R29).

Fig. 3(c) compares the modeling and experimental results for C5-species. The computed mole fraction of $c-C_5H_5$ is higher than the experimental value, probably because its quenching on the nozzle surface causes its measured concentration to be lowered.

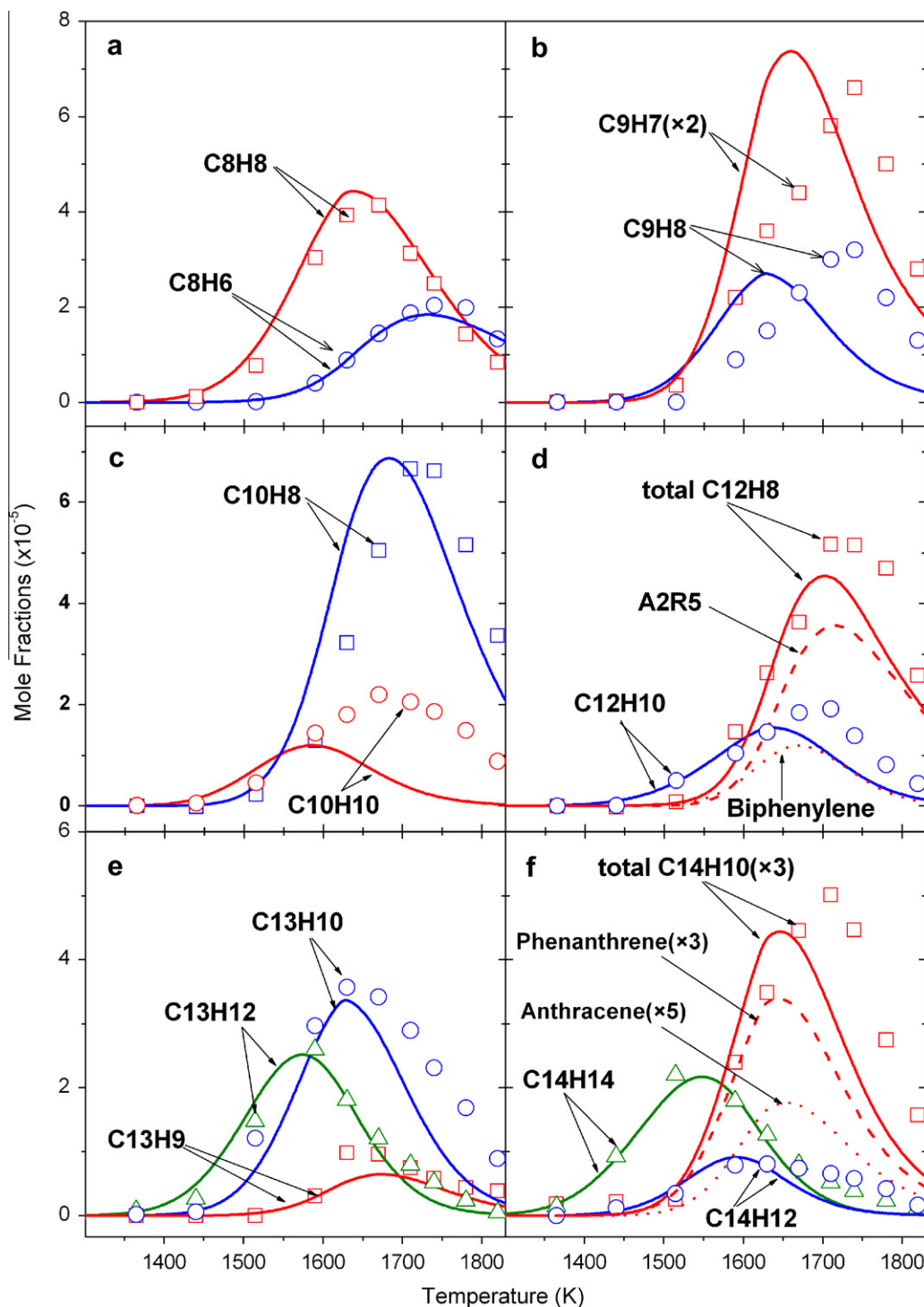


Fig. 4. Comparison between experimental and modeling predictions of mole fraction profiles of aromatic hydrocarbons including C8–C14-species. The experimental data are shown as open symbols. The solid lines denote the modeling results by the model M1. The dash and dot lines are the modeling results of different isomers.

In addition, the mole fraction profiles of C_5H_6 and C_5H_4 are also consistent with the experimental results within the accepted error range. Most of $c-C_5H_5$ comes from the decomposition of the benzyl radical (R9, 29% contribution) and $C_7H_6 + H = c-C_5H_5 + C_2H_2$ pathway (R10, 71% contribution). $c-C_5H_5$ decomposes rapidly to yield acetylene and the propargyl radical via the ring-opening reaction (R41). This reaction contributes about 53% to the formation of C_3H_3 , while the decomposition of the phenyl radical (R24) contributes 18% and the decomposition of C_7H_5 accounts for 17% on the production of C_3H_3 . The propargyl radical is a significant interme-

diate in the pyrolysis of toluene and the growth of PAHs. As showed in Fig. 3(d), the predicted X_M of C_3H_3 is 2.27×10^{-3} at about 1710 K, while the experimental X_M is 4.40×10^{-4} at the same temperature. The experimental X_M is lower than the modeling result, probably due to the quenching effect. Fig. 3(d) also presents the comparison between experimental and computed results of C_4H_4 , $a-C_3H_4$ and $p-C_3H_4$. The predicted levels and shapes of $a-C_3H_4$ and $p-C_3H_4$ are consistent with the experimental results. Reaction flux analysis (Tables 3 and 4) shows that most of the $p-C_3H_4$ and $a-C_3H_4$ are produced from C_3H_3 (R53 and R54,

Fig. 2 presents the comparison between experimental and simulated results for the major products (C_2H_2) and polyynes (C_{2n}H_2 , $n=2-5$) mentioned above. The concentrations of these species increase monotonically as the temperature increases, which can be

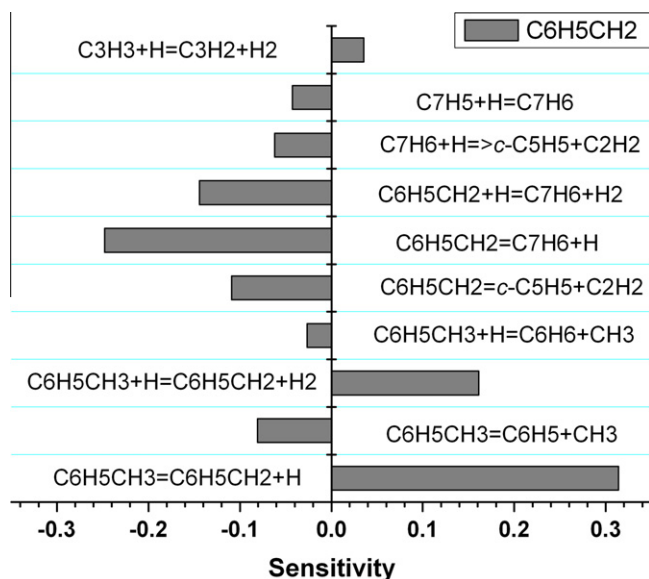


Fig. 7. Sensitivity analysis of conversion of the benzyl radical at 1670 K. Only the reactions with an absolute value of the sensitivity coefficient higher than 0.02 are considered.

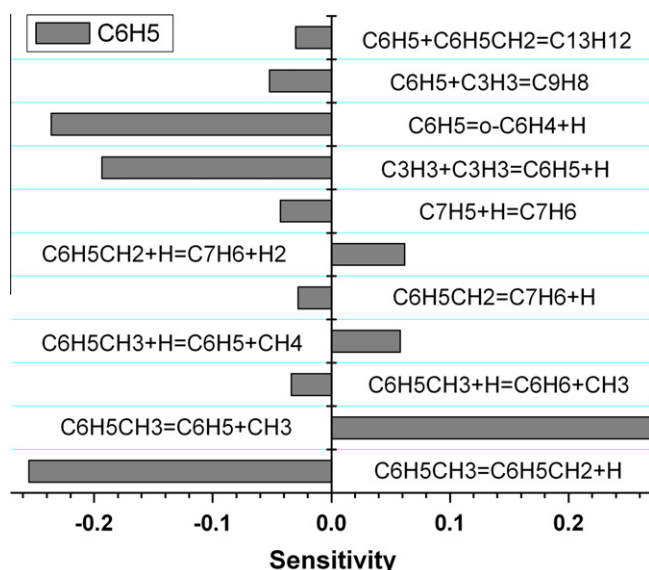


Fig. 8. Sensitivity analysis of conversion of the phenyl radical at 1670 K. Only the reactions with an absolute value of the sensitivity coefficient higher than 0.02 are considered.

explained by their growing thermodynamic stability with the increasing temperature reported by Krestinin [82]. C_2H_2 and polyynes decompose only at higher temperatures. In the case of C_2H_2 , decomposition occurs at temperatures higher than 2000 K [82]. In the experiment, the measured mole fractions decrease with increasing carbon numbers, from 6.33×10^{-3} (C_2H_2) to 8.26×10^{-6} ($C_{10}H_2$) at 1820 K [34]. Both experimental and computed T_{Form} of C_2H_2 are around 1470 K, which is lower than those of polyynes. Acetylene is produced mainly by five reactions including $c-C_5H_5 = C_3H_3 + C_2H_2$ (R41, 34% contribution), $C_7H_6 + H = c-C_5H_5 + C_2H_2$ (R10, 25%), $C_6H_5CH_2 = c-C_5H_5 + C_2H_2$ (R9, 10%), $C_7H_5 + H = C_5H_4 + C_2H_2$ (R14, 8%), and $C_5H_4 + H = C_3H_3 + C_2H_2$ (R42, 6%). Sensitivity analysis for acetylene shows that the reaction $C_6H_5CH_3 = C_6H_5CH_2 + H$ (R1) is also important for its formation, as seen in Fig. 7. In addition, the reactions $C_7H_6 + H \rightarrow c-C_5H_5 + C_2H_2$ (R10),

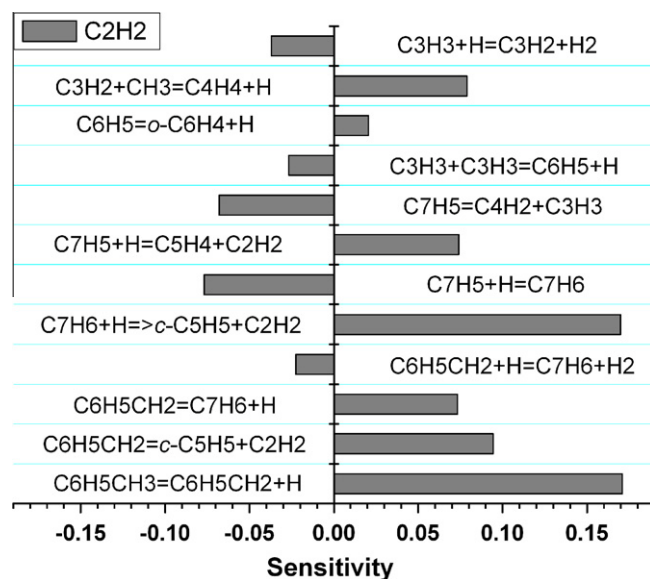


Fig. 9. Sensitivity analysis of conversion of acetylene at 1670 K. Only the reactions with an absolute value of the sensitivity coefficient higher than 0.02 are considered.

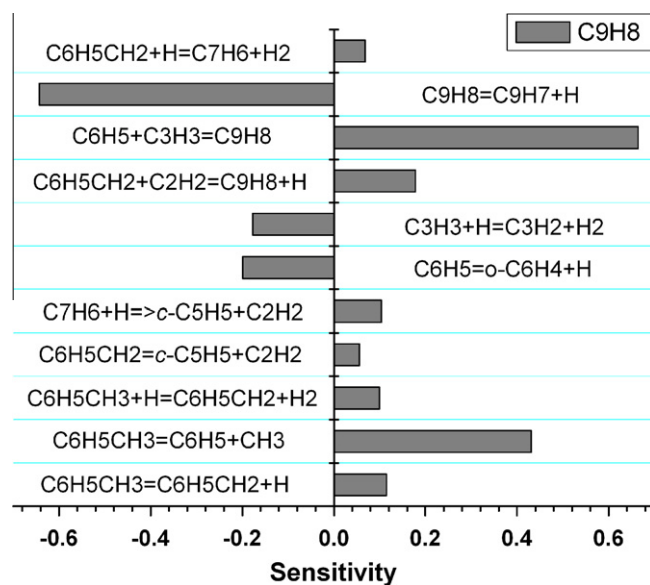


Fig. 10. Sensitivity analysis of conversion of indene at 1670 K. Only the reactions with an absolute value of the sensitivity coefficient higher than 0.02 are considered.

$C_6H_5CH_2 = C_7H_6 + H$ (R7) and $C_6H_5CH_2 = c-C_5H_5 + C_2H_2$ (R9) have inhibiting effects on acetylene formation.

Based on the above discussion, the decomposition pathways of toluene are drawn in Fig. 5. It is concluded that toluene is consumed mainly through the reaction sequence $C_6H_5CH_3 \rightarrow C_6H_5CH_2 \rightarrow C_7H_6 \rightarrow c-C_5H_5 \rightarrow C_3H_3$. This sequence also contributes to most acetylene formation.

3.2.3. PAHs

The experimental and computed mole fraction profiles of PAHs are illustrated in Fig. 4, including C8–C14 species such as phenylacetylene (C_8H_6), styrene (C_8H_8), indenyl (C_9H_7), indene (C_9H_8), naphthalene ($C_{10}H_8$), 1,2-dihydronaphthalene ($C_{10}H_{10}$), acenaphthylene (A2R5, $C_{12}H_8$), biphenylene ($C_{12}H_8$), biphenyl (P2, $C_{12}H_{10}$), fluorene ($C_{13}H_{10}$), diphenylmethane ($C_{13}H_{12}$), phenanthrene ($C_{14}H_{10}$), (Z)-stilbene ($C_{14}H_{12}$) and bibenzyl ($C_{14}H_{14}$). These

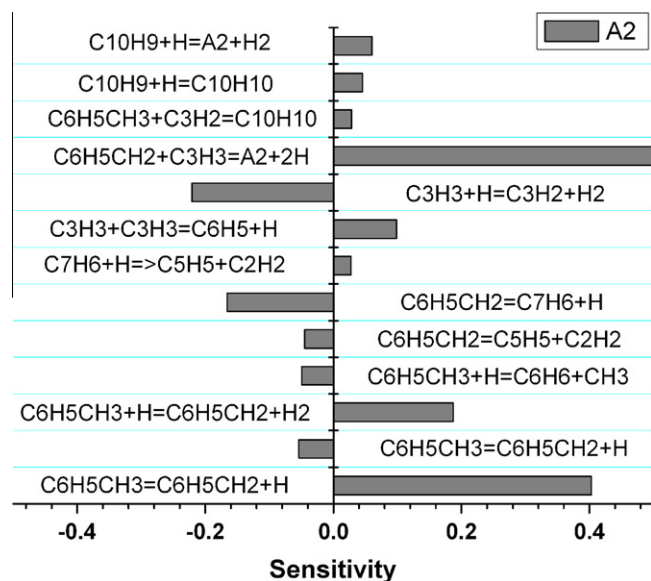


Fig. 11. Sensitivity analysis of conversion of naphthalene at 1670 K. Only the reactions with an absolute value of the sensitivity coefficient higher than 0.02 are considered.

species have bell-shaped mole fraction profiles with X_M ranging from 1×10^{-5} to 1×10^{-4} . The initial formation temperatures are in the range of 1370–1570 K, while the T_M s are in the range of 1520–1770 K. Characteristic reaction pathways for the formation of PAHs are illustrated in Fig. 6, while the contributions of selected species are listed in Table 3. The reactions between the primary products ($C_6H_5CH_2$, C_6H_6 , and C_6H_5) of toluene pyrolysis and C2- and C3-species (C_2H , C_2H_2 , C_3H_2 , C_3H_3) are the initial steps of PAH formation.

Indene is the smallest cyclopentafused PAH (CP-PAH), containing both five- and six-member rings. The simulated peak mole fraction of indene is 2.90×10^{-5} , only 9% smaller than experimental X_M (3.2×10^{-5}) though the calculated T_M is shifted toward low temperatures. Formation channels of indene have been computationally investigated by Vereecken et al. in detail [83,84]. They studied the formation pathway of indene including the reactions between benzyl and acetylene (R67), and phenyl and allene (R68). Kislov and Mebel [85,86] suggested that the reactions between phenyl/benzene and the propargyl radical were also possible formation pathways of indene (R69) and (R70). Recently, da Silva and Bozzelli [87] calculated the indene formation pathway via the reaction of fulvenallene with acetylene (R66). The reactions are adopted in the current model.



As showed in Table 3, indene is produced mainly from the reactions of benzyl (R67, 18% contribution) and phenyl (R69, 81%), while reactions R66, R68 and R70 are less important. The sensitivity analysis of the formation and consumption of indene is shown in Fig. 10. The combination reaction of C_6H_5 and C_3H_3 (R69) shows great importance to the formation of indene. In addition, the reactions related to the formation of benzyl (R1) and (R3) and phenyl (R2) radicals also present promoting effects on indene formation. The most important pathway of indene consumption is the H scission yielding the indenyl radical (R71).

Meanwhile the consumption reactions of propargyl (R52) and phenyl radicals (R25) show negative sensitivity coefficients (SC) for the production of indene.

H scission from indene (R71) leads to the formation of the indenyl radical. Around 44% of the indenyl radical reacts with C_3H_3 and C_3H_2 to form acenaphthylene. This indicates that a small CP-PAH can grow to large CP-PAH through the reactions with C3-species. Computed mole fraction profiles of acenaphthylene (dash line) and its isomer biphenylene (dot line) are shown in Fig. 4(d), respectively. The solid line is the computed profile of total $C_{12}H_8$. As seen, the predicted results are in good agreement with the experimental data.

Concentration profile is provided for the bicyclic aromatic naphthalene ($C_{10}H_8$) in Fig. 4(c). The experimental X_M of naphthalene is 6.65×10^{-5} at 1710 K [33]. The computed mole fraction profile of naphthalene has a similar X_M , but the shape shifts slightly towards low temperatures compared with the experimental results. Naphthalene has been investigated by many experiments and calculations [42,88–95]. It can be seen from Table 2 that naphthalene is produced by many different channels including the reactions between C5- and C5-species (R79), C6- and C4-species (R80, R81 and R82), C7- and C3-species (R78), C8- and C2-species (R84 and R85) and and C9- and C1-species (R83). Reaction flux analysis shows that most of the naphthalene (85%) produced from the reactions between the benzyl radical and C_3H_3 (R78), while about 10% is contributed by the reaction sequences $C_{10}H_{10} \rightarrow C_{10}H_9 \rightarrow C_{10}H_8$. The reactions $C_6H_5CH_2 + C_3H_3 = C_{10}H_8 + H_2$ (R78) exhibit relatively large sensitivity with respect to the formation of naphthalene, as shown in Fig. 11. The formation reactions of benzyl (R1) and (R3) are also important for the formation of naphthalene. While the reactions of consumption of $C_6H_5CH_2$ (R7) and (R9) and C_3H_3 (R52) have an inhibiting effect.

Both diphenylmethane ($C_{13}H_{12}$) and fluorene ($C_{13}H_{10}$) were detected in the experiment [33]. In the model, the combination of phenyl and benzyl radicals yields $C_{13}H_{12}$ (R98), 78% of which is subsequently converted to $C_{13}H_{10}$ through sequential H-abstraction reactions (R99 and R101). The dominant formation channel of phenanthrene ($C_{14}H_{10}$) is similar to that of $C_{13}H_{12}$: two benzyl radicals combine to produce bibenzyl ($C_{14}H_{14}$, R104), which eliminates H_2 to yield stilbene ($C_{14}H_{12}$, R105), and then further converts to phenanthrene via H_2 elimination (R106). About 23% of phenanthrene isomerizes to anthracene through R108. As shown in Fig. 4e and f, computed mole fraction profiles of C13- and C14-species are in good agreement with both experimental X_M and T_M values.

4. Conclusion

A detailed kinetic model consisting of 137 species and 530 elementary reactions was developed to simulate toluene pyrolysis at low pressure and over the temperature range of 1270–1870 K. Satisfactory agreement is obtained between experimental and modeling results including decomposition products and PAHs up to phenanthrene. Reaction flux and sensitivity analysis presented the main decomposition pathways of toluene and the growth of PAHs. Toluene decomposes mainly through the reaction sequence $C_6H_5CH_3 \rightarrow C_6H_5CH_2 \rightarrow C_7H_6 \rightarrow c-C_5H_5 \rightarrow C_3H_3$ which also has a predominant contribution to the production of acetylene. While the phenyl radical, the benzyl radical, C2- and C3-species are critical to the formation of C9- and C10-species, such as indene and naphthalene in the toluene pyrolysis.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2010.04.002](https://doi.org/10.1016/j.combustflame.2010.04.002).

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